

RESEARCH PAPER
NASSAU CANDY DISTRIBUTOR
SALES, PROFITABILITY & BUSINESS ANALYTICS

An End-to-End Applied Business Analytics Study

Academic / Internship Project Submission
2026

Research Paper Information

Project title: Nassau Candy Distributor – Sales, Profitability & Business Analytics

Study type: Applied descriptive business analytics case study

Dataset: Nassau Candy Distributor transaction dataset

Records: 10,194 | Products: 15 | Divisions: 3

Primary outputs: Excel workbook, PDF report, validated dataset and Streamlit dashboard

Declaration

Submitted by: GAURI S. SHEWALE

I declare that this project report documents an applied analytics study using the supplied Nassau Candy Distributor dataset and the analytical outputs created for this project.

The numerical results are based on the validated project dataset. External methodological sources are cited in the references section.

This report is intended for academic and internship evaluation and is not an audited financial statement.

Certificate / Verification Page

Field	Information
Submitted by	GAURI S. SHEWALE
Project	Nassau Candy Distributor – Sales, Profitability & Business Analytics
Study type	Applied descriptive business analytics case study
Dataset	Nassau Candy Distributor transaction dataset
Year	2026
Deliverables	Excel workbook, research report, validated dataset and Streamlit dashboard
Field	Entry
S	t
—	—

Acknowledgement

This project provided an opportunity to apply data cleaning, financial validation, exploratory analysis, spreadsheet modeling, visualization and dashboard development to a practical distributor-sales dataset.

The work also benefited from established analytics methodology literature, particularly the CRISP-DM process and research discussing its use in business analytics.

Abstract

This study investigates sales performance, profitability, cost structure and margin risk in a Nassau Candy Distributor transaction dataset. An end-to-end descriptive analytics workflow was applied, beginning with data preparation and validation and continuing through exploratory analysis, division and product profitability, Pareto concentration, cost and margin diagnostics, visualization and dashboard development.

The validated dataset contains 10,194 records, total sales of 141,783.63, total cost of 48,340.83, gross profit of 93,442.80 and 38,654 units. The weighted gross margin is 65.91%. Chocolate contributes 92.88% of revenue and 95.06% of gross profit. Kazookles has the lowest product margin at 7.69%.

Keywords: business analytics, profitability, gross margin, Pareto analysis, data quality, Python, Excel, Streamlit.

Executive Summary

KPI	Value
R	e
1	0

The project transforms transaction records into a management-oriented view of revenue, cost, profit, product concentration and margin risk.

Table of Contents

No.	Contents	Page
1	Introduction	9
1.1	Business Context	10
1.2	Importance of Data Quality	11
2	Research Problem and Objectives	12
2.1	Research Questions	13
3	Literature Review	14
3.1	Literature Review – Descriptive Analytics	15
4	Dataset Description	16
4.1	Dataset Scope	17
5	Research Methodology	18
5.1	Analytical Workflow	19
6	Data Cleaning and Quality Validation	20
6.1	Data Quality	21
6.2	Derived Metrics	22
7	Exploratory Data Analysis	23
7.1	Overall Interpretation	24
8	Division Performance Analysis	25
8.1	Division Margin Interpretation	26
8.2	Division Cost Diagnostics	27
9	Product Profitability Analysis	28
9.1	Product Profitability Categories	29
9.2	Highest-Margin Product	30
9.3	Margin-Risk Product	31
10	Pareto Analysis	32
10.1	Revenue Pareto Interpretation	33
10.2	Profit Pareto Interpretation	34
10.3	Pareto Management Implications	35
11	Cost and Margin Analysis	36
11.1	Company Margin Benchmark	37
11.2	Cost Efficiency Categories	38
11.3	Pricing Inefficiency	39
11.4	Cost Inefficiency	40
12	Dashboard and Visualization Design	41
12.1	Excel Workbook	42
12.2	Streamlit Dashboard	43
12.3	Dashboard Preparation	44

No.	Contents	Page
13	Findings and Discussion	45
13.1	Scale versus Margin	46
13.2	Portfolio Concentration	47
14	Business Recommendations	48
14.1	Action Matrix	49
15	Limitations	50
15.1	Reproducibility	51
16	Conclusion	52
17	Future Scope	53
18	References	54
19	Appendices	55
Appendix A	Data Dictionary	56
Appendix B	KPI Formulas	57
Appendix C	Product Summary	58
Appendix D	Division Summary	59
Appendix E	Streamlit Run Guide	60
Appendix F	Submission Checklist	61

1. Introduction

2. Research Problem and Objectives

3. Literature Review

4. Dataset Description

Field	Description	Purpose
Row ID	Record identifier	Record tracking
Order ID	Order reference	Order grouping
Order Date	Transaction date	Time analysis
Ship Date	Recorded shipment date	Date validation
Ship Mode	Shipment method	Operational context
Customer ID	Customer identifier	Customer analysis
Division	Business/product division	Division analysis
Region	Geographic region	Regional analysis
Product ID	Product identifier	Product grouping
Product Name	Product description	Product analysis
Sales	Revenue value	Revenue KPI
Units	Units sold	Volume analysis
Gross Profit	Sales minus Cost	Profit KPI

Field	Description	Purpose
Cost	Recorded product cost	Cost analysis

5. Research Methodology

Stage	Project Implementation
Business understanding	Define sales, profitability, cost and margin questions
Data understanding	Inspect fields, ranges, missing values and anomalies
Data preparation	Normalize text, validate financial fields and flag duplicates/date issues
Descriptive analysis	Calculate KPIs and division/product summaries
Diagnostic analysis	Perform Pareto, cost and margin-risk analysis
Visualization	Build Excel charts and Streamlit dashboard
Evaluation	Check calculations, consistency and interpretation
Delivery	Prepare workbook, report, validated dataset and application

6. Data Cleaning and Quality Validation

Check	Result	Status
Records	10,194	Validated
Duplicate records flagged	620 across 307 groups	Retained for transparency
Invalid/missing Sales	0	Passed
Invalid/missing Cost	0	Passed
Invalid/missing Gross Profit	0	Passed
Negative financial values	0	Passed
Gross Profit mismatch	0	Passed
Invalid/zero Units	0	Passed
Missing Product/Division	0	Passed
Shipping-date anomaly	10,194 records	Flagged for investigation

7. Exploratory Data Analysis

8. Division Performance

9. Product Profitability

10. Pareto Analysis

Measure	Result	Interpretation
Revenue Pareto	5 of 15 products = 33.33%	First five exceed 80% cumulative revenue
Revenue boundary	92.883008%	Above 80% threshold
Profit Pareto	5 of 15 products = 33.33%	First five exceed 80% cumulative profit
Profit boundary	95.057747%	Above 80% threshold

11. Cost and Margin Analysis

Division	Cost % of Sales	Gross Margin	Diagnostic	Interpretation
Chocolate	32.55%	67.45%	Cost efficient	Margin above benchmark
Other	55.16%	44.84%	Cost pressure	Margin below benchmark
Sugar	33.39%	66.61%	Cost efficient	Small revenue base

12. Dashboard Design

13. Findings and Discussion

14. Recommendations

15. Limitations

16. Conclusion

17. Future Scope

18. References

Author / Organization	Year	Research / Resource	Publisher / Venue	DOI / Source
Jaggia, S.; Kelly, A.; Lertwachara, K.; Chen, L.	2020	Applying the CRISP-DM Framework for Teaching Business Analytics	Decision Sciences Journal of Innovative Education, Wiley	10.1111/dsji.12222
Mariscal, G.; Marbán, Ó.; Fernández, C.	2021	A Systematic Literature Review on Applying CRISP-DM Process Model	Procedia Computer Science, Elsevier	10.1016/j.procs.2021.01.199
Python Software Foundation	2026	Python Documentation	Python Software Foundation	python.org/doc
pandas Development Team	2026	pandas Documentation	pandas project / NumFOCUS	pandas.pydata.org/docs
Streamlit	2026	Streamlit Documentation	Streamlit, Inc.	docs.streamlit.io
Microsoft	2026	Microsoft Excel Support	Microsoft	support.microsoft.com/excel

19. Appendices

1. Introduction

Business analytics converts operational data into evidence that supports decisions. Distributor datasets can reveal revenue concentration, product profitability, cost pressure and margin differences.

This project treats the Nassau Candy Distributor dataset as an applied case study. The workflow moves from data understanding and validation to descriptive analysis and business communication.

The study is intentionally descriptive. It does not claim to forecast future outcomes. Instead, it explains historical performance and identifies areas requiring further review.

1.1 Business Context

Distributor businesses operate across products, divisions, customers, geography and transaction periods. Management therefore needs summarized information that preserves the relationship between sales, cost, volume and profit.

A high total margin can hide weak products, while a high-margin product may contribute little absolute profit if its sales volume is small. This project explicitly addresses both dimensions.

1.2 Importance of Data Quality

Analytical results are only as reliable as the inputs and calculations behind them. The project therefore validates Sales, Cost, Gross Profit and Units before creating management KPIs.

Date anomalies and duplicate groups are documented rather than silently corrected. This approach improves transparency and helps reviewers distinguish source-data limitations from analytical findings.

2. Research Problem and Objectives

The central problem is converting raw transaction records into reliable and actionable profitability information.

- Measure overall sales, cost, gross profit and gross margin.
- Compare division-level performance.
- Rank products by gross profit and margin.
- Identify margin-risk products.
- Measure revenue and profit concentration.
- Diagnose cost efficiency.
- Create reusable analytical dashboards.

2.1 Research Questions

- What is the overall profitability profile?
- Which division drives revenue and profit?
- Which products generate the greatest gross profit?
- Which products have unusually low margins?
- How concentrated are revenue and profit?
- What does cost as a percentage of sales reveal?
- How can results be communicated interactively?

3. Literature Review

CRISP-DM is widely used as a structured process for data-mining and analytics projects. Published literature describes phases including business understanding, data understanding, data preparation, modeling, evaluation and deployment.

Jaggia et al. discuss CRISP-DM in business analytics education and emphasize problem formulation, data wrangling, evaluation and storytelling. A systematic review also describes CRISP-DM as a widely adopted process while identifying deployment as an area that can require additional attention.

Recent work extends CRISP-DM toward stronger data quality, governance, traceability and iterative validation. These ideas are relevant to this project because the analysis depends on validated financial fields and documented transformations.

3.1 Literature Review – Descriptive Analytics

For descriptive business analytics, the modeling stage can be interpreted as preparation of analytical queries, aggregations and diagnostic views rather than predictive machine learning.

This project follows that interpretation: the main analytical objects are KPIs, division summaries, product profitability, contribution measures, Pareto curves and margin-risk classifications.

4. Dataset Description

Field Group	Examples	Purpose
I	d	e
R	o	w
T	r	a

The analysis focuses primarily on financial, product and division fields because these directly support the study objectives.

4.1 Dataset Scope

Metric	Result
R	e
1	0

The shipping-date range is materially later than the order-date range and is therefore treated as an important source-data issue.

5. Research Methodology

The project follows a CRISP-DM-inspired workflow adapted for descriptive business analytics.

Stage	Implementation
B	u
D	e

5.1 Analytical Workflow

- CSV ingestion
- Text and column normalization
- Duplicate detection
- Financial validation
- Date validation
- Derived metrics
- EDA
- Division analysis
- Product analysis
- Pareto analysis
- Cost and margin diagnostics
- Dashboard development
- Business recommendations

The workflow is iterative: quality findings can require returning to data preparation before final interpretation.

6. Data Cleaning and Quality Validation

Check	Result
R	e
1	0

All financial records passed the core arithmetic validation. Duplicate groups were flagged for transparency rather than assumed to be erroneous.

6.1 Date Quality

Diagnostic	Finding
O	r
2	0

The date issue should be investigated against the original source system before the dataset is used for operational delivery analysis.

6.2 Derived Metrics

Measure	Formula
G	r
G	r

7. Exploratory Data Analysis

EDA begins with the overall business profile and then decomposes the results by division and product.

KPI	Value
S	a
1	4

7.1 Overall Interpretation

The aggregate margin is strong, but the average alone does not describe the portfolio. Segment analysis is required to identify whether the result is broad-based or concentrated.

The next sections therefore examine division contribution, product profitability and cost burden.

8. Division Performance Analysis

Division	Sales	Cost	Gross Profit	Units	Margin	Revenue %	Profit %
Chocolate	131,692.90	42,868.28	88,824.62	37,275	67.45%	92.88%	95.06%
Other	9,663.25	5,329.80	4,333.45	1,242	44.84%	6.82%	4.64%
Sugar	427.48	142.75	284.73	137	66.61%	0.30%	0.30%
Division	Sales	Cost	GP	Units	Margin	Revenue %	Profit %
C	h	o	c	o	l	a	t
1	3	1	,	6	9	2	.
4	2	,	8	6	8	.	2
8	8	,	8	2	4	.	6
3	7	,	2	7	5		
6	7	.	4	5	%		
9	2	.	8	8	%		
9	5	.	0	6	%		

Chocolate is the dominant economic engine. Other has the lowest margin and cost efficiency, while Sugar has a strong percentage margin but a very small scale.

8.1 Division Margin Interpretation

Chocolate's 67.45% margin is above the company benchmark of 65.91%. Other's 44.84% margin is substantially below it. Sugar's 66.61% margin is also above the benchmark.

This demonstrates why both margin and contribution should be monitored. A high margin on a tiny revenue base does not necessarily produce a large amount of total profit.

8.2 Division Cost Diagnostics

Division	Cost % Sales	Margin	Diagnostic
C	h	o	c
3	2	.	5
6	7	.	4
C	o	s	t

The Other division is the clearest division-level cost diagnostic target.

9. Product Profitability Analysis

Product	Sales	Gross Profit	Units	Margin
Wonka Bar -Scrumdiddlyumptious	27,874.80	19,357.50	7,743	69.44%
Wonka Bar - Triple Dazzle Caramel	28,485.00	18,610.20	7,596	65.33%
Wonka Bar - Milk Chocolate	26,867.75	17,443.37	8,267	64.92%
Wonka Bar - Nutty Crunch Surprise	23,574.95	16,819.95	6,755	71.35%
Wonka Bar - Fudge Mallows	24,890.40	16,593.60	6,914	66.67%
Lickable Wallpaper	7,860.00	3,930.00	393	50.00%
Wonka Gum	597.50	310.70	478	52.00%
Everlasting Gobstopper	130.00	104.00	13	80.00%
Kazookles	1,205.75	92.75	371	7.69%
Hair Toffee	76.50	59.50	17	77.78%
Product	Sales	Gross Profit	Units	Margin
W	o	n	k	a
2	8	,	4	8
1	8	,	6	1
7	,	5	9	6
6	5	.	3	3

9.1 Product Profitability Categories

Category	Margin Rule
E	X
>	=

The project uses these categories as descriptive labels. The overall benchmark is used separately for company-level margin risk.

9.2 Highest-Margin Product

Everlasting Gobstopper has the highest observed margin at 80.00%, but its sales are only 130.00. Hair Toffee also has a very high margin but a very small revenue base.

Wonka Bar - Nutty Crunch Surprise combines a high margin of 71.35% with substantial sales and gross profit, making it a more economically significant high-margin product.

9.3 Margin-Risk Product

Kazookles has a gross margin of 7.69%, making it the principal margin-risk product in the analysis.

Metric	Kazookles	Company Benchmark
S	a	1
1	,	2
1	4	1

10. Pareto Analysis

Pareto analysis measures concentration by ranking products from largest to smallest contribution and calculating cumulative percentages.

Measure	Finding
P	r
1	5

10.1 Revenue Pareto Interpretation

The first five products account for more than 80% of revenue. This indicates a concentrated revenue portfolio and means availability, pricing and demand for these products can materially affect total results.

10.2 Profit Pareto Interpretation

The first five products account for approximately 95.06% of gross profit. Profit concentration is therefore very high and reinforces the need to monitor the core products closely.

10.3 Pareto Management Implications

- Protect availability of core products.
- Monitor pricing and discounts on high-contribution products.
- Review low-margin products separately from low-volume products.
- Refresh Pareto analysis periodically.
- Use Pareto as a prioritization tool, not as an automatic discontinuation rule.

11. Cost and Margin Analysis

Cost and margin analysis explains the relationship between revenue and the resources consumed to generate it.

Measure	Purpose
C	o
M	e

11.1 Company Margin Benchmark

The weighted company gross margin is 65.91%. This is the benchmark used for the Watch/Low margin-risk classification.

The benchmark should be interpreted as a portfolio-level reference rather than an external industry standard.

11.2 Cost Efficiency Categories

Category	Rule
V	e
>	6

These thresholds are analytical heuristics created for dashboard prioritization and are not accounting standards.

11.3 Pricing Inefficiency

Diagnostic	Rule
H	i
M	a

Pricing diagnostics should be validated against customer contracts, discounts and market prices before action.

11.4 Cost Inefficiency

Diagnostic	Rule
C	r
>	6

12. Dashboard and Visualization Design

Dashboard Component	Prepared Value / Definition
Data source	Validated Nassau Candy Distributor dataset
Records	10,194
Products / Divisions	15 / 3
Sales	141,783.63
Cost	48,340.83
Gross Profit	93,442.80
Units	38,654
Weighted Gross Margin	65.91%
Core visuals	KPI cards, division comparison, product profitability, Sales vs Margin, Pareto
Risk view	Margin-risk product and cost-efficiency diagnostics
Technology	Excel, Python/Pandas, Streamlit
Visual	Purpose
K	P
S	a

12.1 Excel Workbook

The final Excel workbook provides the structured analytical artifact. It contains an executive dashboard, division performance, product profitability, Pareto analysis, margin risk, data quality, validated data and final findings.

The workbook is intended to allow reviewers to move from management-level KPIs to detailed product-level evidence.

12.2 Streamlit Dashboard

The Streamlit application provides an interactive interface for KPI review, division analysis, product profitability, sales-versus-margin analysis and Pareto information.

Pandas handles data processing, NumPy supports numerical operations, and Streamlit provides the web interface.

13. Findings and Discussion

Finding	Evidence	Meaning
S	t	r
6	5	.
H	i	g

13.1 Scale versus Margin

The analysis demonstrates that absolute contribution and percentage margin must be considered together. A small product can have an exceptional margin without being a major profit driver.

Conversely, a large product with a slightly lower margin can produce more gross profit because of its sales scale.

13.2 Portfolio Concentration

The strong concentration of revenue and profit creates an efficiency advantage because management can focus on a small group of economically important products.

The same concentration can create dependency risk if demand, supply or pricing changes materially for those products.

14. Business Recommendations

- Protect availability and monitor pricing for core products.
- Investigate Kazookles supplier cost, selling price, discounts and product configuration.
- Analyze the Other division at product level to identify procurement or pricing pressure.
- Refresh Pareto analysis regularly.
- Validate shipping-date logic before operational delivery analysis.

14.1 Action Matrix

Area	Issue	Suggested Review
K	a	z
7	.	6
P	r	i

15. Limitations

- Historical dataset may not represent current conditions.
- Analysis is descriptive, not predictive.
- No external market-price or supplier data were supplied.
- Shipping dates contain a major anomaly.
- Duplicate groups require source-system confirmation.
- Risk thresholds are analytical heuristics.
- No causal claims are established.

15.1 Reproducibility

Reproducibility is supported by the validated CSV, Excel workbook, Python application and requirements file.

Validation	Status
S	a
P	a

16. Conclusion

This project demonstrates an end-to-end applied analytics workflow for the Nassau Candy Distributor dataset. It combines data quality, descriptive analysis, profitability diagnostics, Pareto concentration and interactive reporting.

The overall gross margin is 65.91%. Chocolate dominates the business, while Other has lower margin and higher cost burden. Kazookles is the principal low-margin exception at 7.69%.

The results show how validated transaction data can be transformed into structured business intelligence that supports management review and prioritization.

17. Future Scope

- Monthly and quarterly trend analysis after date validation.
- Customer profitability.
- Regional profitability.
- Inventory integration.
- Supplier-cost integration.
- Sales and profit forecasting.
- Anomaly detection.
- Price/cost scenario analysis.
- Automated dashboard refresh.

18. References

- Jaggia, S., Kelly, A., Lertwachara, K., & Chen, L. (2020). Applying the CRISP-DM Framework for Teaching Business Analytics. *Decision Sciences Journal of Innovative Education*, 18(4), 612–634. <https://doi.org/10.1111/dsji.12222>
- Mariscal, G., Marbán, Ó., & Fernández, C. (2021). A Systematic Literature Review on Applying CRISP-DM Process Model. *Procedia Computer Science*, 181, 526–534. <https://doi.org/10.1016/j.procs.2021.01.199>
- Python Software Foundation. Python Documentation. <https://www.python.org/doc/>
- Pandas Development Team. pandas Documentation. <https://pandas.pydata.org/docs/>
- Streamlit. Documentation. <https://docs.streamlit.io/>
- Microsoft. Excel Support. <https://support.microsoft.com/excel>

Appendix A – Data Dictionary

Field	Description
R	o
U	n

Appendix B – KPI Formulas

KPI	Formula
T	o
S	U

Appendix C – Product Summary

Product	Sales	GP	Units	Margin
W	0	n	k	a
2	7	,	8	7
1	9	,	3	5
7	,	7	4	3
6	9	.	4	4

Appendix D – Division Summary

Division	Sales	Cost	GP	Units	Margin
C	h	o	c	o	l
1	3	1	,	6	9
4	2	,	8	6	8
8	8	,	8	2	4
3	7	,	2	7	5
6	7	.	4	5	%

Appendix E – Streamlit Run Guide

- Install Python.
- Open the project folder.
- Run pip install -r requirements.txt.
- Run streamlit run streamlit_app.py.
- Upload the validated CSV if prompted.
- Review KPIs, division and product analysis.
- Use the dashboard outputs for demonstration.

Appendix F – Submission Checklist

Deliverable	File
E	x
N	a

Final Research Note

This document is a detailed applied analytics research-style paper. It is not presented as a peer-reviewed publication. If submitted to a journal or conference, the author information, formatting and reference style should be adapted to the venue's requirements.

For internship submission, the paper should be submitted together with the analytical workbook, validated dataset and dashboard so that the results can be reviewed and reproduced.

END OF RESEARCH PAPER

Research Paper Status and Data Accuracy

This document is a formal research-style project report prepared from the supplied Nassau Candy Distributor dataset. It is not represented as a peer-reviewed journal article, published paper, or externally approved research publication.

All principal numerical findings in this report are derived from the validated project dataset: 10,194 records; Sales 141,783.63; Cost 48,340.83; Gross Profit 93,442.80; Units 38,654; and weighted Gross Margin 65.91%.

A material source-data issue was identified in the recorded dates. Order dates range from 2024-01-02 to 2025-12-31, while Ship Dates range from 2026-06-30 to 2030-06-28. The median calculated shipping delay is 1,274 days and all 10,194 records exceed 30 days. The dates were therefore flagged rather than silently replaced.

The report should be submitted with the final Excel workbook, validated dataset and Streamlit source/application files where those materials are required by the submission instructions.